

Paper 1 - Set 11 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	C	Indices and surds
2	D	Simultaneous equations and inequalities
3	B	Circles and circle theorems
4	D	Integration
5	F	Ratio, proportion and percentages
6	A	Differentiation
7	A	Number, units and estimation
8	F	Sequences and series
9	B	Quadratic functions
10	E	Indices and surds
11	D	Circles and circle theorems
12	C	Sequences and series
13	C	Quadratic functions
14	D	Angles, polygons, congruence and vectors
15	D	Binomial expansion
16	A	Polynomials, factor and remainder theorems
17	D	Triangles, radians and sectors
18	F	Trigonometric functions and equations
19	F	Integration
20	A	Number, units and estimation

Worked solutions

Question 1 IND-15 • Indices and surds • spec MM1.1

Given that $2^x = 5$ and $2^y = 20$, find the value of 2^{2x-y} .

Answer: C

$$2^{2x-y} = \frac{(2^x)^2}{2^y} = \frac{25}{20} = \frac{5}{4}.$$

Question 2 SIM-07 • Simultaneous equations and inequalities • spec MM1.5

The inequality $x^2 + bx + 9 > 0$ holds for all real x . Find the complete set of values of b .

Answer: D

Need discriminant negative: $b^2 - 36 < 0$, so $-6 < b < 6$.

Question 3 CIR-13 • Circles and circle theorems • spec MM3.2

The circle $x^2 + y^2 = r^2$ and the line $3x + 4y = 20$ have exactly one point in common. Find r .

Answer: B

Distance from origin to the line $= \frac{20}{\sqrt{9+16}} = 4$. Tangency means $r = 4$.

Question 4 INT-18 • Integration • spec MM7.3

Let $I = \int_0^1 (x^2 + 1) dx$ and $J = \int_0^1 (x + 1)^2 dx$. Which of the following is true?

Answer: D

$$I = \frac{1}{3} + 1 = \frac{4}{3}, J = \int_0^1 (x^2 + 2x + 1) dx = \frac{1}{3} + 1 + 1 = \frac{7}{3}, J - I = 1.$$

Question 5 RAT-05 • Ratio, proportion and percentages • spec M3.11

A bacteria population doubles every 3 hours. By what factor does it grow in 24 hours?

Answer: F

$$24/3 = 8 \text{ doublings: } 2^8 = 256.$$

Question 6 DIF-12 • Differentiation • spec MM6.3

The line $y = 4x + c$ is a tangent to $y = x^2 + 2x + 5$. Find c .

Answer: A

$$2x + 2 = 4 \Rightarrow x = 1, y = 8. 8 = 4 + c \Rightarrow c = 4.$$

Question 7 NUM-20 • Number, units and estimation • spec M2.11

Find the exact value of $(\sqrt{5} - 1)^2 + (\sqrt{5} + 1)^2$.

Answer: A

$$(6 - 2\sqrt{5}) + (6 + 2\sqrt{5}) = 12.$$

Question 8 SEQ-18 · Sequences and series · spec MM2.1

A sequence is defined by $u_1 = 1$ and $u_{n+1} = u_n + 2n + 1$. Which of the following is an expression for u_n ?

Answer: F

$$u_2 = 4, u_3 = 9, u_4 = 16; u_n = n^2. \text{ Check: } (n+1)^2 = n^2 + 2n + 1.$$

Question 9 QUA-23 · Quadratic functions · spec MM1.3

The line $y = mx$ meets the curve $y = x^2 + 9$ at two distinct points. Find the complete set of values of m .

Answer: B

$$x^2 - mx + 9 = 0 \text{ needs } m^2 - 36 > 0, \text{ so } m < -6 \text{ or } m > 6.$$

Question 10 IND-05 · Indices and surds · spec MM1.2

Simplify $\sqrt{75} - \sqrt{27} + \sqrt{12}$.

Answer: E

$$\sqrt{75} = 5\sqrt{3}, \sqrt{27} = 3\sqrt{3}, \sqrt{12} = 2\sqrt{3}, \text{ so the total is } (5 - 3 + 2)\sqrt{3} = 4\sqrt{3}.$$

Question 11 CIR-12 · Circles and circle theorems · spec MM3.2

The circle C has equation $x^2 + y^2 - 4x - 6y + 9 = 0$. Which of the following lines does not meet C ?

Answer: D

Centre $(2, 3)$, radius 2, so the circle spans $0 \leq x \leq 4$ and $1 \leq y \leq 5$. The line $y = 0$ is 3 from the centre, more than the radius, so it misses; $x = 0$, $x = 4$ and $y = 5$ touch and $x = 2$ passes through the centre.

Question 12 SEQ-13 · Sequences and series · spec MM2.2

An arithmetic sequence has $u_3 = 10$ and $u_8 = 25$. Find the smallest value of n for which $u_n > 100$.

Answer: C

$$5d = 15 \Rightarrow d = 3, a = 4. u_n = 4 + 3(n-1) = 3n + 1 > 100 \Rightarrow n > 33, \text{ so } n = 34.$$

Question 13 QUA-08 · Quadratic functions · spec MM1.3

The equation $kx^2 + 4x + k = 0$ has no real roots. Find the complete set of values of k .

Answer: C

Need $16 - 4k^2 < 0$, i.e. $k^2 > 4$, so $k < -2$ or $k > 2$. (If $k = 0$ the equation is linear with one root, so $k = 0$ is excluded anyway.)

Question 14 GEO-05 · Angles, polygons, congruence and vectors · spec M5.5

Two similar triangles have corresponding sides of length 6 cm and 9 cm. The smaller triangle has area 20 cm². Find the area of the larger triangle.

Answer: D

$$\text{Area scale factor } \left(\frac{9}{6}\right)^2 = \frac{9}{4}: 20 \times \frac{9}{4} = 45.$$

Question 15 BIN-08 · Binomial expansion · spec MM2.4

The first three terms in the expansion of $(1 + ax)^n$ in ascending powers of x are $1 + 12x + 54x^2$. Find n .

Answer: D

$na = 12$ and $\frac{n(n-1)}{2}a^2 = 54$. Dividing the second by the square of the first: $\frac{n-1}{2n} = \frac{54}{144} = \frac{3}{8}$, so $8n - 8 = 6n$ and $n = 4$ (then $a = 3$; check $\binom{4}{2} \cdot 9 = 54$).

Question 16 POL-10 · Polynomials, factor and remainder theorems · spec MM1.6

Given that $x^3 + px + q$ is divisible by $(x - 2)^2$, find the value of q .

Answer: A

$x^3 + px + q = (x - 2)^2(x - r)$. Comparing x^2 coefficients: $-4 - r = 0 \Rightarrow r = -4$. Constant term: $(4)(4) = 16 = q$. (Also $p = 4 + 4r = -12$.)

Question 17 TRI-15 · Triangles, radians and sectors · spec MM4.1

Triangle ABC has $AB = 7$, $BC = 8$, $CA = 9$. Find the area of the triangle.

Answer: D

$\cos A = \frac{49 + 81 - 64}{2 \cdot 7 \cdot 9} = \frac{66}{126} = \frac{11}{21}$, so $\sin A = \frac{\sqrt{441 - 121}}{21} = \frac{\sqrt{320}}{21} = \frac{8\sqrt{5}}{21}$. Area $= \frac{1}{2} \cdot 7 \cdot 9 \cdot \frac{8\sqrt{5}}{21} = 12\sqrt{5}$.

Question 18 TRG-14 · Trigonometric functions and equations · spec MM4.6

Solve $12\cos^2 x + 6\sin x - 10 = 2$ for $0^\circ < x < 360^\circ$. Which of the following is the complete set of solutions?

Answer: F

$12(1 - \sin^2 x) + 6\sin x - 12 = 0 \Rightarrow 12\sin^2 x - 6\sin x = 0 \Rightarrow 6\sin x(2\sin x - 1) = 0$. $\sin x = 0$ gives 180° (the endpoints 0° and 360° are excluded); $\sin x = \frac{1}{2}$ gives 30° and 150° .

Question 19 INT-01 · Integration · spec MM7.2

Evaluate $\int_1^4 \frac{3x^2 - 5}{\sqrt{x}} dx$.

Answer: F

Integrand $= 3x^{3/2} - 5x^{-1/2}$; integral $= \left[\frac{6}{5}x^{5/2} - 10x^{1/2} \right]_1^4 = \left(\frac{6 \cdot 32}{5} - 20 \right) - \left(\frac{6}{5} - 10 \right) = \frac{192}{5} - 20 - \frac{6}{5} + 10 = \frac{186}{5} - 10 = \frac{136}{5}$.

Question 20 NUM-09 · Number, units and estimation · spec M2.3

How many positive divisors does 360 have?

Answer: A

$360 = 2^3 \cdot 3^2 \cdot 5$, so the number of divisors is $(3 + 1)(2 + 1)(1 + 1) = 24$.